

TrES-5b

R-0730

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250522 · created 2026-07-28T00:53:09+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460817.8723 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.052
Transit depth (Rp/Rs)^2	3.41 +/- 1.9 [%]
Orbital Inclination (inc)	89.64 +/- 2.52
Airmass coefficient 1 (a1)	0.579 +/- 0.014
Airmass coefficient 2 (a2)	0.47 +/- 0.18
Scatter in the residuals of the lightcurve fit is	2.49 %
Best Comparison Star	#3 - [290, 252]
Optimal Aperture	3.26
Optimal Annulus	9.6
Transit Duration (day)	0.0346 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.185 $\pm$ 0.052 vs 0.142 (deviation 0.83 σ)
Duration fit vs published	0.0346 d vs 0.0758 d (deviation 8.41 σ)
Mid-time O-C	-10.2 min
Depth SNR	3.6
Residual scatter	2.49 %

Light curve

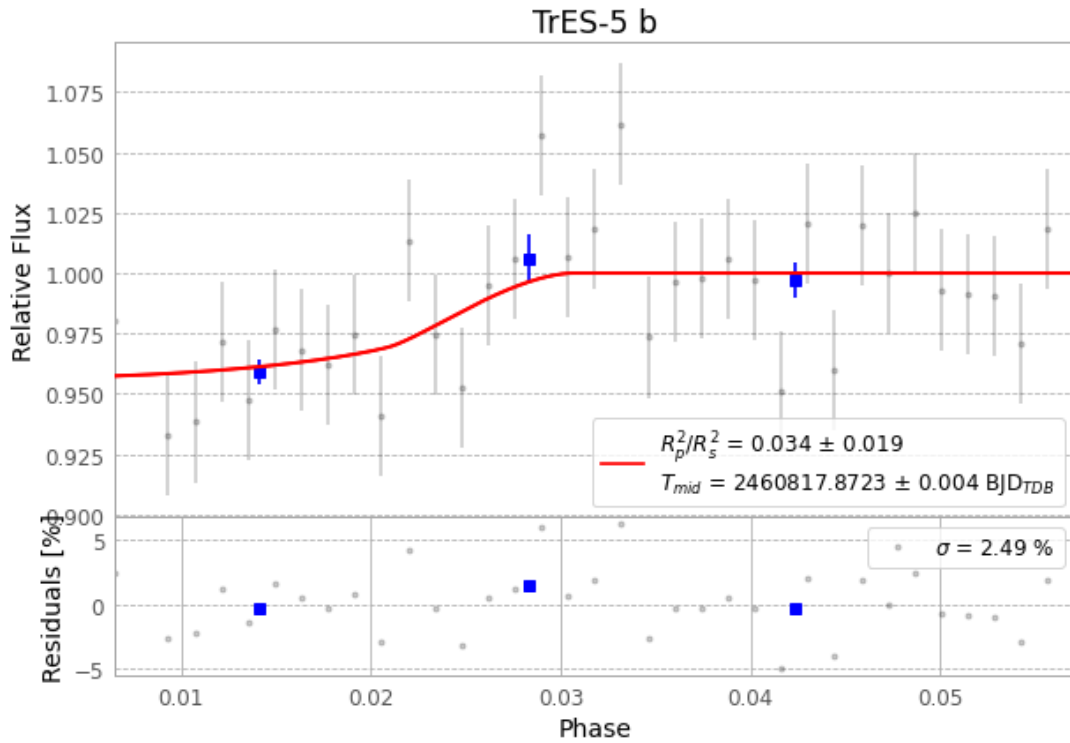


Figure 1 — FinalLightCurve_TrES-5 b_2025-05-22.png (SHA-256 a4c693fbff2c947f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eef7f9dfaf2f6e8d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

36 FITS frames (24 MB), TRES-5250522090915.FITS → TRES-5250522105715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250522.log (4403ae262bfd...), FinalLightCurve_TrES-5 b_2025-05-22.png (a4c693fbff2c...), FinalLightCurve_TrES-5 b_2025-05-22.csv (89af4d4634bd...)

Compute resources

Wall time 100.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 00:53Z.